

Comprehensive PYQ Analysis & Exam Blueprint Report

Basic Civil Engineering & Engineering Mechanics (BT-204 GS)

University: Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), MP

Course Code: BT-204 (GS)

Analysis Horizon: 2022 – 2025 (8 Exam Cycles)

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Section 1: Unit-Wise Previous Year Questions (PYQs)

This section compiles every question extracted from the 8 consecutive semesters (2022–2025) without omission, sorted logically into standard RGPV syllabus units.

Unit I: Building Materials

Write on the characteristics of concrete's chemical and physical state.

Dec 2024

Nov 2022

Explain the concrete preparation procedure in detail.

Dec 2024

Briefly discuss the compaction and curing process of concrete.

Dec 2024

Explain various methods of compaction of concrete and their importance.

Dec 2025

Explain various laboratory tests on concrete and mortar with neat sketches.

Dec 2025

June 2024

June 2022

Define workability of the Concrete and discuss the factors that affect it.

June 2024

Dec 2023

June 2023

June 2022

Explain nominal and design mix concrete in detail.

Dec 2023

Define any two-field tests that are conducted on cement and sand.

Dec 2023

What is the composition of cement? Explain in detail.

Dec 2023

Define the following terms: Initial and final setting time of cement, Efflorescence, Curing of concrete.

Dec 2023

Define the following terms with respect to concrete: Segregation, Bleeding, Proportioning.

June 2023

Define any two-laboratory tests conducted on bricks and cement.

June 2023

What are the different characteristics of stones and their methods of testing? **June 2025**

Write the names of various types of stones which are used in building work and construction industry.

June 2025 **June 2024**

Narrate the classification of stones used in construction industry. **Nov 2022**

What is mortar? Describe the estimation of mortar requirement. **June 2025**

What are the constituents, desirable properties, and types of Paints? **June 2025**

Describe the manufacturing process of bricks. Explain the properties of good building bricks and laboratory tests conducted on bricks. **Dec 2025**

Unit II: Building Construction

Describe in brief about the different types of RCC footing / shallow foundations. **Dec 2024** **June 2024**

Dec 2023 **June 2023** **June 2022**

Describe different types of staircases with neat sketches and explain their suitability for different buildings.

Dec 2025 **June 2024** **Dec 2023** **Nov 2022**

Discuss the general principles that need to be observed in brick masonry construction. **June 2025**

Explain various types of stone masonry. Draw typical sketches to illustrate them. **June 2025**

Describe various types of windows and doors used in building construction. **June 2025** **Dec 2023**

Define and explain Plastering and Pointing. **Dec 2023** **June 2023**

What is the purpose to provide the foundation and describe its various types in detail? **June 2023**

List out the various stages for the construction of a building and building materials used. **June 2023**

What are the various types of bonds used in the construction of brick masonry? **June 2023** **June 2022**

Explain the terms: Roofs, Curing of masonry. **June 2022**

Unit III: Surveying & Levelling

Explain the different methods of measuring elevations, including spirit leveling, trigonometric leveling, and barometric leveling. **Dec 2025**

What are electronic surveying instruments (EDM)? Explain their need and advantages over conventional instruments. **Dec 2025** **June 2024** **Dec 2023** **June 2023**

Describe the steps involved in preparing a profile and cross-section for a proposed road project. **Dec 2025**
June 2024 **June 2022** **Nov 2022**

Define contour. Explain the characteristics/properties of contour lines and their significance/contour interval/ maps. **Dec 2025** **June 2025** **Dec 2023** **June 2023** **Nov 2022**

Explain the conventional methods of distance measurement using chains and tapes, and errors in chain surveying. **Dec 2025** **June 2025** **June 2023**

Define remote sensing. Describe components and applications in terrain analysis, material inventories, site investigation, and building industry. **Dec 2025** **June 2025** **Dec 2024** **June 2024** **Dec 2023** **June 2023**
Nov 2022

With a neat sketch, explain the principle of positioning with GPS. **June 2025**

Explain with a neat sketch: (i) Traversing by included angles, (ii) Traversing by direct angles. **June 2025**

Use a sketch to illustrate each Theodolite component. **Dec 2024** **Nov 2022**

Explain in brief about Reciprocal Levelling. **Dec 2024** **June 2023**

Explain the 'auto level' and 'dumpy level' with neat sketches. **June 2023** **June 2022**

What is local attraction in surveying? How is it detected and corrected? Define True and Magnetic Bearings, Fore Sight and Back Sight. **Dec 2023** **June 2023**

Numerical (Levelling - Rise & Fall): Entering readings and computing RLs with arithmetic checks.

Dec 2024

June 2024

June 2022

Nov 2022

Numerical (Area & Volume): Earthwork calculations by Trapezoidal/Prismoidal methods and boundaries by Mid-ordinate, Average-ordinate, Trapezoidal, Simpson's Rules.

Dec 2024

June 2024

June 2022

Unit IV: Engineering Mechanics: Force Systems, Trusses & Beams

Explain the procedure to find forces in members of a truss by using the method of sections and method of joints.

Dec 2025

Dec 2024

June 2024

June 2023

Nov 2022

State and derive/prove Lami's Theorem and solve related equilibrium problems.

June 2024

Nov 2022

Write down the steps for finding the resultant of a concurrent coplanar force system.

June 2025

June 2023

June 2022

State D'Alembert's principle and its application.

June 2025

Dec 2023

State the assumptions used in simple bending and derive the bending equation.

June 2025

Define support reactions and their types. Explain and provide the procedure to draw SFD and BMD for simply supported and cantilever beams under point loads, UDL, UVL, and couples.

Dec 2024

Dec 2023

June 2023

June 2022

Nov 2022

Numerical (Truss Analysis): Determine the forces in all/specific members under various loading configurations.

Dec 2025

Dec 2024

June 2022

Numerical (Beam Reactions/SFD/BMD): Solve cantilever or simply supported beams with mixed loading profiles.

Dec 2025

Dec 2023

June 2023

June 2022

Unit V: Centroid, Centre of Gravity, Moment of Inertia & Friction

Determine the centre of gravity of a solid cone of base radius 'R' and height 'h'.

Dec 2025

Define centroid and centre of gravity. Explain the differences between them with suitable examples.

Dec 2025

June 2024

Nov 2022

Define and derive an expression for the Parallel Axis Theorem.

Dec 2023

Differentiate the difference between Moment of Inertia of mass, Moment of Inertia of Area, and Product of Inertia.

June 2024

Nov 2022

Briefly explain Principle Axes and Product of Inertia with examples.

Dec 2024

Numerical (Moment of Inertia): Find MI for standard or composite laminae (H-shape, T-shape, or shaded shapes with circular cutouts).

June 2024

Dec 2023

June 2023

Nov 2022

Numerical (Friction & Blocks): Find the minimum effort P to cause motion left/right for systems of blocks on horizontal or inclined surfaces.

Dec 2024

June 2022

Nov 2022

State the Work-Energy Principle. Solve the rolling uniform cylinder kinetic energy question.

June 2025

Determine the diameter and total elongation of a 200m vertical hanging rod under axial pull.

June 2025

Section 2: Important Topics & Priority Matrix

This matrix outlines the core concepts evaluated in RGPV exam cycles, weighed by frequency and structured for high-yield preparation.

Unit	Topic / Concept	Priority	Exam Relevance / Reasoning
Unit I	Concrete Workability & factors affecting it; Lab/Field tests on Cement, Concrete & Bricks.	★★★★ High	Appears in almost every session; crucial for core evaluation.
Unit II	Staircase types & suitability; Footing types/Foundations.	★★★★ High	High-scoring sketch questions recurring consistently.
Unit III	Remote Sensing components & applications; Levelling numerical calculations (Rise & Fall).	★★★★ High	Guaranteed 7-14 marks across all cycles.
Unit IV	Truss Analysis (Method of Joints/Sections); Beam Support Reactions, SFD & BMD.	★★★★ High	Heavyweight numericals requiring structured practice.
Unit V	Moment of Inertia of composite sections (T, H, Shaded laminae); Parallel axis theorem.	★★★★ High	Predictable high-scoring composite lamina problems.
Unit III	Contour characteristics & properties; Area/Volume Earthwork computations.	★★★ Medium	Tested frequently as an alternative or optional question.
Unit IV	Lami's Theorem proof & numerical application; D'Alembert's principle.	★★★ Medium	Provides straightforward mathematical marks.
Unit V	Friction block systems on horizontal/ inclined planes.	★★★ Medium	Tests analytical equilibrium modeling skills.
Unit I	Stone characteristics & names; Paints composition & properties.	★ Low	Descriptive theory tested occasionally.

Section 3: Most Repeated PYQs & Trend Analysis

The following questions represent identical or closely matched core problems that repeat across multiple exam cycles.

1. Remote Sensing Principles, Components, and Applications

- **Asked in:** Dec 2025, June 2025, Dec 2024, June 2024, Dec 2023, June 2023, Nov 2022
- **Frequency:** 7 times (Highest Recurring Theory Topic)

2. Workability of Concrete and Influencing Factors

- **Asked in:** June 2024, Dec 2023, June 2023, June 2022
- **Frequency:** 4 times

3. Types of Staircases with Sketches and Suitability

- **Asked in:** Dec 2025, June 2024, Dec 2023, Nov 2022
- **Frequency:** 4 times

4. Numerical: Levelling Field Book Entries via Rise and Fall Method

- **Asked in:** Dec 2024, June 2024, June 2022, Nov 2022
- **Frequency:** 4 times (Highest Recurring Surveying Numerical)

5. Support Reactions, Shear Force & Bending Moment Diagrams (SFD & BMD)

- **Asked in:** Dec 2025, Dec 2024, Dec 2023, June 2023, June 2022, Nov 2022
- **Frequency:** 6 times

6. Analysis of Plane Truss (Method of Joints vs Method of Sections)

- **Asked in:** Dec 2025, Dec 2024, June 2024, June 2023, June 2022, Nov 2022
- **Frequency:** 6 times

7. Moment of Inertia of Shaded or Composite Lamina (T, H, or Cutout shapes)

- **Asked in:** June 2024, Dec 2023, June 2023, Nov 2022
- **Frequency:** 4 times